

Sedentary time in adults and the association with diabetes, cardiovascular disease and death: systematic review and meta-analysis.



AIMS/HYPOTHESIS: Sedentary (sitting) behaviours are ubiquitous in modern society. We conducted a systematic review and meta-analysis to examine the association of sedentary time with diabetes, cardiovascular disease and cardiovascular and all-cause mortality. METHODS: Medline, Embase and the Cochrane Library databases were searched for terms related to sedentary time and health outcomes. Cross-sectional and prospective studies were included. RR/HR and 95% CIs were extracted by two independent reviewers. Data were adjusted for baseline event rate and pooled using a random-effects model. Bayesian predictive effects and intervals were calculated to indicate the variance in outcomes that would be expected if new studies were conducted in the future. RESULTS: Eighteen studies (16 prospective, two cross-sectional) were included, with 794,577 participants. Fifteen of these studies were moderate to high quality. The greatest sedentary time compared with the lowest was associated with a 112% increase in the RR of diabetes (RR 2.12; 95% credible interval [CrI] 1.61, 2.78), a 147% increase in the RR of cardiovascular events (RR 2.47; 95% CI 1.44, 4.24), a 90% increase in the risk of cardiovascular mortality (HR 1.90; 95% CrI 1.36, 2.66) and a 49% increase in the risk of all-cause mortality (HR 1.49; 95% CrI 1.14, 2.03). The predictive effects and intervals were only significant for diabetes. CONCLUSIONS/INTERPRETATION: Sedentary time is associated with an increased risk of diabetes, cardiovascular disease and cardiovascular and all-cause mortality; the strength of the association is most consistent for diabetes.